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LECTURE NOTES ON ABSTRACT ALGEBRA

Module 1: Groups, Subgroups & Cosets — Module 2: Normal Subgroups, Quotient Groups & Homomorphisms

Course Code: MATMJ51 — Reference Text: Abstract Algebra (Dummit & Foote) — Platform: www.sciqb.com

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1 Module 1: Groups, Subgroups, and Fundamental Examples

Lecture Dates: August 11 – 12, 2026

Abstract algebra studies algebraic structures such as groups, rings, and fields. A group is one of the most fundamental algebraic structures, capturing the formal properties of symmetry, transformations, and algebraic operations.

1.1 Definition of a Group and Axioms

Definition: Group

Let G be a non-empty set and let $*$ be a binary operation defined on G . The pair $(G, *)$ is called a **group** if it satisfies the following four fundamental axioms:

1. **Closure Axiom:**

$$\forall a, b \in G, \quad a * b \in G \quad (1)$$

2. **Associative Axiom:**

$$\forall a, b, c \in G, \quad (a * b) * c = a * (b * c) \quad (2)$$

3. **Existence of Identity Element:**

$$\exists e \in G \text{ such that } \forall a \in G, \quad a * e = e * a = a \quad (3)$$

4. **Existence of Inverse Element:**

$$\forall a \in G, \quad \exists a^{-1} \in G \text{ such that } a * a^{-1} = a^{-1} * a = e \quad (4)$$

If, in addition, the binary operation satisfies the **Commutative Property** ($\forall a, b \in G, a * b = b * a$), then $(G, *)$ is called an **Abelian (or Commutative) Group**.

1.2 Standard Examples of Groups

The following collection represents standard examples of infinite and finite groups:

Fundamental Examples of Groups

1. **Real Numbers under Addition:** $(\mathbb{R}, +)$ with identity element $e = 0$ and inverse of a given by $-a$.
2. **Non-zero Real Numbers under Multiplication:** $(\mathbb{R}^*, \cdot)$ where $\mathbb{R}^* = \mathbb{R} \setminus \{0\}$, identity $e = 1$, and inverse $a^{-1} = 1/a$.
3. **Integers under Addition:** $(\mathbb{Z}, +)$ with identity $e = 0$.
4. **Non-zero Complex Numbers under Multiplication:** $(\mathbb{C}^*, \cdot)$ where $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$.
5. **Integers Modulo n under Addition:** $(\mathbb{Z}_n, +)$ where $\mathbb{Z}_n = \{\bar{0}, \bar{1}, \bar{2}, \dots, \overline{n-1}\}$.
6. **Rational Numbers under Addition:** $(\mathbb{Q}, +)$.
7. **Matrix Group under Addition:** $(M_n(\mathbb{R}), +)$, the set of all $n \times n$ matrices over $\mathbb{R}$ under matrix addition.
8. **Klein 4-Group (V_4):** $V_4 = \{e, a, b, c\}$ where $c = ab$, satisfying $a^2 = b^2 = c^2 = e$ and $ab = ba = c, ac = ca = b, bc = cb = a$.
9. **Polynomial Group under Addition:** $(P_n(\mathbb{R}), +)$, polynomials of degree $\leq n$ with real coefficients.
10. **Non-zero Rational Numbers under Multiplication:** $(\mathbb{Q}^*, \cdot)$.
11. **Quaternion Group (Q_8):** $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ under quaternion multiplication, where $i^2 = j^2 = k^2 = ijk = -1$.
12. **Symmetric Group ($S_n, \circ$):** Permutation group on n symbols.
13. **Group of Units Modulo n ($U(n), \cdot$):** $U(n) = \{k \in \mathbb{Z}_n \mid \gcd(k, n) = 1\}$ under multiplication modulo n .

1.3 Symmetric Groups (S_n) and Permutation Analysis

Let $T_n = \{1, 2, 3, \dots, n\}$ be a set of n distinct elements.

Combinatorial Properties of Mapping Set T_n

- **Cardinality of Set:** $|T_n| = n$.
- **Total Number of Functions $f : T_n \rightarrow T_n$:** n^n .
- **Total Number of One-to-One (Injective) Functions:** $n!$.
- **Total Number of Bijective Functions:** $n!$.

Definition of Symmetric Group: The set $S_n = \{f : T_n \rightarrow T_n \mid f \text{ is bijective}\}$ under function composition $\circ$ forms a non-abelian group (for $n \geq 3$) of order $|S_n| = n!$, called the **Symmetric Group** of degree n .

1.4 Group of n -th Roots of Unity (G_n) and Isomorphism with $\mathbb{Z}_n$

Consider the complex set of n -th roots of unity:

$$G_n = \left\{ e^{\frac{2\pi i k}{n}} \mid 0 \leq k \leq n-1 \right\} = \left\{ \cos\left(\frac{2\pi k}{n}\right) + i \sin\left(\frac{2\pi k}{n}\right) \mid 0 \leq k \leq n-1 \right\} \quad (5)$$

Examples of Roots of Unity Groups

- **For $n = 1$:** $G_1 = \{1\}$ (Trivial group)
- **For $n = 2$:** $G_2 = \{1, -1\}$
- **For $n = 3$:** $G_3 = \{1, \omega, \omega^2\}$ where $\omega = e^{2\pi i/3} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$
- **For $n = 4$:** $G_4 = \{1, -1, i, -i\}$

Define a mapping $f : \mathbb{Z}_n \rightarrow G_n$ by:

$$f(\bar{k}) = e^{\frac{2\pi i k}{n}} \quad (6)$$

Notice that for $\bar{k}_1, \bar{k}_2 \in \mathbb{Z}_n$:

$$f(\bar{k}_1 + \bar{k}_2) = e^{\frac{2\pi i(k_1+k_2)}{n}} = e^{\frac{2\pi i k_1}{n}} \cdot e^{\frac{2\pi i k_2}{n}} = f(\bar{k}_1) \cdot f(\bar{k}_2) \quad (7)$$

Since f is a bijective operation-preserving mapping, $(\mathbb{Z}_n, +)$ is isomorphic to $(G_n, \cdot)$, written $\mathbb{Z}_n \cong G_n$.

2 Module 2: Group Homomorphisms and Isomorphisms

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Homomorphisms are structural mappings between algebraic structures that preserve the underlying operations.

2.1 Formal Definitions

Definition: Group Homomorphism

Let $(G_1, \circ)$ and $(G_2, \cdot)$ be two groups. A mapping $\phi : G_1 \rightarrow G_2$ is called a **group homomorphism** if:

$$\phi(g_1 \circ g_2) = \phi(g_1) \cdot \phi(g_2) \quad \forall g_1, g_2 \in G_1 \quad (8)$$

Definition: Isomorphism and Isomorphic Groups

- A homomorphism $\phi : G_1 \rightarrow G_2$ is called an **isomorphism** if ϕ is both **one-to-one (injective)** and **onto (surjective)**.
- Two groups $(G_1, \circ)$ and $(G_2, \cdot)$ are said to be **isomorphic** (denoted $G_1 \cong G_2$) if there exists an isomorphism $\phi : G_1 \rightarrow G_2$.

3 Module 3: Normal Subgroups and Center of a Group

Lecture Date: August 12, 2026

3.1 Normal Subgroups

Definition: Normal Subgroup

Let G be a group and let H be a subgroup of G ($H \leq G$). We say H is a **normal subgroup** of G (denoted $H \triangleleft G$) if:

$$ghg^{-1} \in H \quad \forall g \in G, \forall h \in H \quad (9)$$

Equivalent condition: $gH = Hg$ for all $g \in G$ (left cosets equal right cosets).

3.2 Center of a Group $Z(G)$

Definition: Center of a Group

The **Center of a Group** G , denoted $Z(G)$, is defined as the set of all elements in G that commute with every element of G :

$$Z(G) = \{g \in G \mid gh = hg, \forall h \in G\} \quad (10)$$

Theorem 3.1: $Z(G)$ is a Subgroup of G

Theorem: Let G be a group. Then $Z(G)$ is a subgroup of G ($Z(G) \leq G$).

Proof:

1. **Non-emptiness:** Since the identity $e \in G$ satisfies $eh = he = h$ for all $h \in G$, $e \in Z(G)$. Thus $Z(G) \neq \emptyset$.
2. **One-Step Subgroup Criterion:** Let $g_1, g_2 \in Z(G)$. Then by definition:

$$g_1h = hg_1 \quad \forall h \in G \quad (11)$$

$$g_2h = hg_2 \quad \forall h \in G \quad (12)$$

From $g_2h = hg_2$, multiplying on the left and right by g_2^{-1} yields:

$$g_2^{-1}(g_2h)g_2^{-1} = g_2^{-1}(hg_2)g_2^{-1} \implies hg_2^{-1} = g_2^{-1}h \quad \forall h \in G \quad (13)$$

Now evaluate $(g_1g_2^{-1})h$:

$$\begin{aligned} (g_1g_2^{-1})h &= g_1(g_2^{-1}h) = g_1(hg_2^{-1}) = (g_1h)g_2^{-1} \\ &= (hg_1)g_2^{-1} = h(g_1g_2^{-1}) \quad \forall h \in G \end{aligned} \quad (14)$$

Therefore, $g_1g_2^{-1} \in Z(G)$, which proves that $Z(G) \leq G$. $\square$

Theorem 3.2: $Z(G)$ is a Normal Subgroup of G

Theorem: Let G be a group. Then $Z(G)$ is a normal subgroup of G ($Z(G) \triangleleft G$).

Proof: Let $h_1 \in Z(G)$ and let $g \in G$. Since $h_1 \in Z(G)$, it commutes with every element of G , so $gh_1 = h_1g$. Now examine gh_1g^{-1} :

$$gh_1g^{-1} = (h_1g)g^{-1} = h_1(gg^{-1}) = h_1e = h_1 \in Z(G) \quad (15)$$

Since $gh_1g^{-1} \in Z(G)$ for all $g \in G$ and $h_1 \in Z(G)$, $Z(G)$ is a normal subgroup of G . $\square$

4 Module 4: Cosets, Equivalence Relations, and Lagrange's Theorem

Lecture Date: August 13, 2026

4.1 Left and Right Cosets

Definition: Cosets

Let G be a group, let H be a subgroup of G , and let $a \in G$.

- **Left Coset of H containing a :**

$$aH = \{ah \mid h \in H\} \quad (16)$$

- **Right Coset of H containing a :**

$$Ha = \{ha \mid h \in H\} \quad (17)$$

Special property: If $a \in H$, then $aH = H = Ha$.

4.2 Equivalence Relation Induced by a Subgroup

Let $H \leq G$. Define a binary relation $\sim$ on G by:

$$a \sim b \iff b^{-1}a \in H \quad (18)$$

Theorem 4.1: Equivalence Relation Proof

Theorem: The relation $\sim$ defined by $a \sim b \iff b^{-1}a \in H$ is an equivalence relation on G .

Proof:

1. **Reflexivity:** For any $a \in G$, $a^{-1}a = e \in H$ (since H is a subgroup). Thus $a \sim a$.

2. **Symmetry:** Suppose $a \sim b \implies b^{-1}a \in H$. Since H is closed under inverses:

$$(b^{-1}a)^{-1} = a^{-1}(b^{-1})^{-1} = a^{-1}b \in H \implies b \sim a \quad (19)$$

3. **Transitivity:** Suppose $a \sim b$ and $b \sim c \implies b^{-1}a \in H$ and $c^{-1}b \in H$. Since H is closed under multiplication:

$$(c^{-1}b)(b^{-1}a) = c^{-1}(bb^{-1})a = c^{-1}ea = c^{-1}a \in H \implies a \sim c \quad (20)$$

Since $\sim$ is reflexive, symmetric, and transitive, it is an equivalence relation on G . $\square$

4.3 Congruence Classes and Partition of G

The equivalence class of an element $a \in G$ under $\sim$ is given by:

$$[a] = \{b \in G \mid a \sim b\} = \{b \in G \mid b^{-1}a \in H\} = aH \quad (21)$$

Key Properties:

- $[a] = [b] \iff a \sim b \iff aH = bH$.
- Two left cosets aH and bH are either identical or disjoint ($aH \cap bH = \emptyset$).
- The distinct left cosets partition G : $G = \bigcup_{aH \in S} aH$.

4.4 Lagrange's Theorem and Orders of Elements

Theorem 4.2: Lagrange's Theorem

Theorem: If G is a finite group and H is a subgroup of G , then the order of H divides the order of G :

$$|H| \mid |G| \quad (22)$$

The ratio $[G : H] = \frac{|G|}{|H|}$ is called the **index** of H in G , representing the number of distinct left cosets.

Proof Outline: Since G is finite, G is partitioned into $k = [G : H]$ distinct disjoint left cosets $a_1H, a_2H, \dots, a_kH$. Because $|a_iH| = |H|$ for each i , we have:

$$|G| = |a_1H| + |a_2H| + \dots + |a_kH| = k|H| \implies |H| \mid |G| \quad \square$$

Corollaries of Lagrange's Theorem

1. **Order of Element Divides Group Order:** For any $a \in G$ in a finite group G , $|a| \mid |G|$, where $|a| = |\langle a \rangle|$.
2. **Groups of Prime Order are Cyclic:** If $|G| = p$ where p is a prime number, then G has no non-trivial subgroups and $G \cong \mathbb{Z}_p$.

5 Module 5: Quotient Groups and Canonical Examples

Lecture Date: August 14, 2026

5.1 Definition and Group Structure of Quotient Groups

Definition: Quotient Group (Factor Group)

Let G be a group and $H \triangleleft G$ be a normal subgroup. The set of left cosets:

$$G/H = \{aH \mid a \in G\} \quad (23)$$

forms a group under the operation of coset multiplication:

$$(aH) \cdot (bH) = (ab)H \quad \forall a, b \in G \quad (24)$$

The group $(G/H, \cdot)$ is called the **Quotient Group** (or Factor Group) of G by H , with order $|G/H| = [G : H] = |G|/|H|$.

5.2 The Canonical Quotient Group $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$

Consider the additive group of integers $(\mathbb{Z}, +)$ and the subgroup $n\mathbb{Z} = \{nk \mid k \in \mathbb{Z}\}$.

- Since $(\mathbb{Z}, +)$ is abelian, every subgroup is normal ($n\mathbb{Z} \triangleleft \mathbb{Z}$).
- The cosets of $n\mathbb{Z}$ in $\mathbb{Z}$ are:

$$\bar{0} = 0 + n\mathbb{Z} = \{\dots, -2n, -n, 0, n, 2n, \dots\} \quad (25)$$

$$\bar{1} = 1 + n\mathbb{Z} = \{\dots, -2n + 1, -n + 1, 1, n + 1, 2n + 1, \dots\} \quad (26)$$

$$\vdots$$

$$\overline{n-1} = (n-1) + n\mathbb{Z} \quad (27)$$

- Thus $\mathbb{Z}/n\mathbb{Z} = \{0 + n\mathbb{Z}, 1 + n\mathbb{Z}, \dots, (n-1) + n\mathbb{Z}\} \cong \mathbb{Z}_n$.

Fact vs Counterexample: Normality in Groups

Fact: In an abelian group, every subgroup is normal.

Counterexample (Non-Normal Subgroup): Consider $G = S_3$ and $H = \{e, (12)\}$. Left coset: $(23)H = \{(23), (23)(12)\} = \{(23), (132)\}$. Right coset: $H(23) = \{(23), (12)(23)\} = \{(23), (123)\}$. Since $(23)H \neq H(23)$, H is **not** a normal subgroup of S_3 .

6 Module 6: Kernels, Images, and Isomorphism Theorems

Lecture Date: August 14, 2026

6.1 Kernel of a Homomorphism $\text{Ker } \phi$

Definition: Kernel

Let $\phi : (G_1, *) \rightarrow (G_2, \circ)$ be a group homomorphism. The **Kernel** of ϕ , denoted $\text{Ker } \phi$, is the set of all elements in G_1 mapped to the identity element $e_2 \in G_2$:

$$\text{Ker } \phi = \{g \in G_1 \mid \phi(g) = e_2\} \quad (28)$$

Fundamental Lemma: Properties of Identity and Inverses under Homomorphism

Lemma: Let $\phi : G_1 \rightarrow G_2$ be a group homomorphism with identities $e_1 \in G_1$ and $e_2 \in G_2$. Then:

1. $\phi(e_1) = e_2$
2. $\phi(g^{-1}) = (\phi(g))^{-1} \quad \forall g \in G_1$

Proof:

1. $\phi(e_1) = \phi(e_1 * e_1) = \phi(e_1) \circ \phi(e_1)$. Multiplying both sides by $[\phi(e_1)]^{-1} \in G_2$:

$$\begin{aligned} e_2 &= [\phi(e_1)]^{-1} \circ \phi(e_1) = [\phi(e_1)]^{-1} \circ (\phi(e_1) \circ \phi(e_1)) \\ &= ([\phi(e_1)]^{-1} \circ \phi(e_1)) \circ \phi(e_1) = e_2 \circ \phi(e_1) = \phi(e_1) \implies \phi(e_1) = e_2 \quad \square \end{aligned} \quad (29)$$

2. $\phi(g) \circ \phi(g^{-1}) = \phi(g * g^{-1}) = \phi(e_1) = e_2$. Multiplying by $(\phi(g))^{-1}$ on the left gives:

$$\phi(g^{-1}) = (\phi(g))^{-1} \quad \square$$

Theorem 6.1: $\text{Ker } \phi$ is a Subgroup of G_1

Theorem: Let $\phi : G_1 \rightarrow G_2$ be a homomorphism. Then $\text{Ker } \phi$ is a subgroup of G_1 ($\text{Ker } \phi \leq G_1$).

Proof:

1. Since $\phi(e_1) = e_2$, $e_1 \in \text{Ker } \phi \implies \text{Ker } \phi \neq \emptyset$.
2. Let $a, b \in \text{Ker } \phi \implies \phi(a) = e_2$ and $\phi(b) = e_2$. Evaluate $\phi(a * b^{-1})$:

$$\begin{aligned} \phi(a * b^{-1}) &= \phi(a) \circ \phi(b^{-1}) = \phi(a) \circ (\phi(b))^{-1} \\ &= e_2 \circ (e_2)^{-1} = e_2 \circ e_2 = e_2 \end{aligned} \quad (30)$$

Thus $a * b^{-1} \in \text{Ker } \phi$. By the subgroup test, $\text{Ker } \phi \leq G_1$. $\square$

Theorem 6.2: Ker ϕ is a Normal Subgroup of G_1

Theorem: Let $\phi : G_1 \rightarrow G_2$ be a homomorphism. Then $\text{Ker } \phi$ is a normal subgroup of G_1 ($\text{Ker } \phi \triangleleft G_1$).

Proof: Let $g \in G_1$ and $h \in \text{Ker } \phi \implies \phi(h) = e_2$. Evaluate $\phi(g * h * g^{-1})$:

$$\begin{aligned}\phi(g * h * g^{-1}) &= \phi(g) \circ \phi(h) \circ \phi(g^{-1}) = \phi(g) \circ e_2 \circ (\phi(g))^{-1} \\ &= \phi(g) \circ (\phi(g))^{-1} = e_2\end{aligned}\quad (31)$$

Since $\phi(g * h * g^{-1}) = e_2$, $g * h * g^{-1} \in \text{Ker } \phi$. Thus $\text{Ker } \phi \triangleleft G_1$. $\square$

6.2 Image of a Homomorphism $\text{Im}(\phi)$ **Definition: Image of Homomorphism**

The **Image** of ϕ , denoted $\text{Im}(\phi)$ or $\phi(G_1)$, is:

$$\text{Im}(\phi) = \{\phi(g) \mid g \in G_1\} \subseteq G_2 \quad (32)$$

Theorem 6.3: $\text{Im}(\phi)$ is a Subgroup of G_2

Theorem: Let $\phi : G_1 \rightarrow G_2$ be a homomorphism. Then $\text{Im}(\phi) \leq G_2$.

Proof: Let $y_1, y_2 \in \text{Im}(\phi) \implies \exists g_1, g_2 \in G_1$ such that $y_1 = \phi(g_1)$ and $y_2 = \phi(g_2)$. Evaluate $y_1 \circ y_2^{-1}$:

$$y_1 \circ y_2^{-1} = \phi(g_1) \circ (\phi(g_2))^{-1} = \phi(g_1) \circ \phi(g_2^{-1}) = \phi(g_1 * g_2^{-1}) \quad (33)$$

Since $g_1 * g_2^{-1} \in G_1$, $y_1 \circ y_2^{-1} \in \text{Im}(\phi)$. Therefore, **Image** $\text{Im}(\phi) \leq G_2$. $\square$

6.3 Injectivity Criterion**Theorem 6.4: One-to-One Characterization via Kernel**

Theorem: A group homomorphism $\phi : G_1 \rightarrow G_2$ is one-to-one (injective) if and only if $\text{Ker } \phi = \{e_1\}$.

Proof:

- ($\implies$) Assume ϕ is one-to-one. Let $g \in \text{Ker } \phi \implies \phi(g) = e_2$. Since $\phi(e_1) = e_2$ and ϕ is injective, $g = e_1$. Thus $\text{Ker } \phi = \{e_1\}$.
- ($\impliedby$) Assume $\text{Ker } \phi = \{e_1\}$. Let $a, b \in G_1$ such that $\phi(a) = \phi(b)$. Then:

$$\phi(a) \circ (\phi(b))^{-1} = e_2 \implies \phi(a * b^{-1}) = e_2 \implies a * b^{-1} \in \text{Ker } \phi \quad (34)$$

Since $\text{Ker } \phi = \{e_1\}$, $a * b^{-1} = e_1 \implies a = b$. Thus ϕ is one-to-one. $\square$

7 Module 7: Master Quick Revision Cheat Sheet

1. Group Axioms, Subgroups, and Cyclic Groups

- **Group Axioms** $(G, *)$:

- Closure: $\forall a, b \in G \implies a * b \in G$
- Associativity: $\forall a, b, c \in G \implies (a * b) * c = a * (b * c)$
- Identity: $\exists e \in G$ such that $a * e = e * a = a, \forall a \in G$
- Inverse: $\forall a \in G, \exists a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$
- Abelian (Commutative): $\forall a, b \in G \implies a * b = b * a$

- **One-Step Subgroup Criterion:** A non-empty subset $H \subseteq G$ is a subgroup ($H \leq G$) if and only if $\forall a, b \in H \implies a * b^{-1} \in H$.

- **Order of an Element & Cyclic Groups:**

- Order $|a| = \min\{n \in \mathbb{N} \mid a^n = e\}$. If no such n exists, $|a| = \infty$.
- Cyclic Group: $G = \langle a \rangle = \{a^k \mid k \in \mathbb{Z}\}$.
- Order relation: $|a| = |\langle a \rangle|$.
- If $|a| = n$, then $|a^k| = \frac{n}{\gcd(n, k)}$. The number of generators of a finite cyclic group of order n is $\phi(n)$ (Euler's totient function).
- Every subgroup of a cyclic group is cyclic.

2. Permutations, Cosets, and Lagrange's Theorem

- **Symmetric Group S_n & Alternating Group A_n :**

- Order of S_n is $|S_n| = n!$.
- Disjoint cycles commute: $(a_1 \dots a_k)(b_1 \dots b_m) = (b_1 \dots b_m)(a_1 \dots a_k)$.
- Every permutation is a product of 2-cycles (transpositions).
- Parity is invariant: A permutation is even/odd depending on the parity of transpositions.
- Alternating group A_n consists of all even permutations, with $|A_n| = \frac{n!}{2}$.

- **Cosets & Index:**

- Left coset: $aH = \{ah \mid h \in H\}$; Right coset: $Ha = \{ha \mid h \in H\}$.
- Coset partition: Cosets partition G into mutually disjoint subsets of equal cardinality $|H|$.
- Index $[G : H]$: Number of distinct left (or right) cosets of H in G .

- **Lagrange's Theorem:** If G is a finite group and $H \leq G$, then $|H|$ divides $|G|$, and:

$$|G| = [G : H] \cdot |H|$$

- **Key Corollaries of Lagrange's Theorem:**

- For any $a \in G$, the order $|a|$ divides $|G|$, and $a^{|G|} = e$.
- Every group of prime order p is cyclic and hence abelian ($G \cong \mathbb{Z}_p$).
- Fermat's Little Theorem: If p is prime and $\gcd(a, p) = 1$, then $a^{p-1} \equiv 1 \pmod{p}$.

3. Normal Subgroups, Quotient Groups, and Homomorphisms

- **Normal Subgroup Characterizations ($N \triangleleft G$):**

- $gNg^{-1} = N \quad \forall g \in G$ (or $gng^{-1} \in N \quad \forall g \in G, n \in N$).
- Left and right cosets coincide: $gN = Ng \quad \forall g \in G$.
- Every subgroup of index 2 is normal ($[G : N] = 2 \implies N \triangleleft G$).
- In an abelian group, every subgroup is normal.

- **Quotient (Factor) Group G/N :**

- If $N \triangleleft G$, the set of cosets $G/N = \{gN \mid g \in G\}$ forms a group under the operation $(aN)(bN) = (ab)N$.
- Identity of G/N is $eN = N$; Inverse is $(aN)^{-1} = a^{-1}N$.
- Order: $|G/N| = [G : N] = \frac{|G|}{|N|}$.

- **Homomorphisms & Isomorphism Theorems:**

- Homomorphism $\phi : G_1 \rightarrow G_2$ preserves operation: $\phi(a * b) = \phi(a) \circ \phi(b)$.
- Kernel: $\text{Ker } \phi = \{g \in G_1 \mid \phi(g) = e_2\} \triangleleft G_1$ (Always a normal subgroup!).
- Image: $\text{Im } \phi = \{\phi(g) \mid g \in G_1\} \leq G_2$ (Subgroup of G_2).
- Injectivity Test: ϕ is one-to-one (injective) $\iff \text{Ker } \phi = \{e_1\}$.
- **First Isomorphism Theorem (Fundamental Homomorphism Theorem):**

$$G_1 / \text{Ker } \phi \cong \text{Im } \phi$$

If ϕ is surjective, then $G_1 / \text{Ker } \phi \cong G_2$.